

Technology Push or Market Pull? Strategic Orientation in Business Model Design and Digital Start-up Performance*

Hai Guo , Chao Wang, Zhongfeng Su , and Donghan Wang

The digital revolution is fundamentally reshaping the business landscape by generating abundant new opportunities and novel business models. As a new source of value creation, a start-up's business model is critical for its survival and growth. How, then, should a digital start-up design its business model? The present study departs from existing viewpoints on the topic and proposes a strategic orientation view. By integrating the resource-based view and the demand-side perspective, we examine the impact of strategic orientation in business model design on the performance of start-ups in the digital economy. Using a data set of digital start-ups, our empirical results show that both technology and consumer orientations are beneficial to the performance of start-ups. However, it would be counterproductive for a digital start-up to seek a balance between both strategic orientations in business model design. Furthermore, the positive relationship between consumer orientation and firm performance becomes more prominent in a highly open technological environment, but is weakened in environments characterized by high user interactivity. This study advances business model design research by proposing a strategic orientation view, and contributes to business model theory building by offering an integrated view that embraces the resource-based view and the demand-side perspective.

Introduction

In the last 20 years, the business model has become a salient issue in the fields of strategy, innovation, and entrepreneurship (Foss and Saebi, 2017; Snihur and Zott, 2020; Teece, 2010). Reflecting the logic of value creation and value capture (Casadesus-Masanell and Ricart, 2010; Teece, 2010), the business model is particularly important for start-up companies, because it represents a key mechanism to transform the entrepreneurial opportunities into superior firm performance (Amit and Zott, 2001; George and Bock, 2011; Guo, Tang, Su, and Katz, 2017). Therefore, scholars have endeavored to understand how entrepreneurs effectively design business models (e.g., McDonald and Eisenhardt, 2020; Snihur and Zott, 2020) and how business model design contributes to the growth of start-ups (e.g., Loon and Chik, 2019; Pati, Nandakumar, Ghobadian, Ireland, and O'Regan, 2018; Zott and Amit, 2007).

Research on business model design appears to follow two different streams. The evolutionary perspective considers business model design as a process of experimentation, learning, and adaptation, by addressing how a start-up's business model emerges (McDonald and Eisenhardt, 2020; Snihur and Zott, 2020), and how a firm changes, transforms, or innovates its business model (Aspara, Lamberg, Laukia, and Tikkanen, 2013; Berends, Smith, Reymen, and Podoyntsina, 2016; Sosna, Treviño-Rodríguez, and Velamuri, 2010). Meanwhile, the thematic-based perspective highlights the central idea of business model design, by focusing on design themes that describe the sources of value creation (e.g., Amit and Zott, 2001; Kulins, Leonardy, and Weber, 2016; Pati et al., 2018; Zott and Amit, 2007, 2008, 2010).

A new generation of research suggests that the digital revolution is profoundly reshaping the way of doing business (Amit and Han, 2017; Kohli and Melville, 2019; Rindfleisch, O'Hern, and Sachdev, 2017; Teece, 2018). The business environment has become highly open (Chesbrough and Appleyard, 2007) and interconnected (Nambisan, 2002; Priem, Wenzel, and Koch, 2018), making the business less bounded and predefined (Nambisan, 2017). Both theory and practice reveal that two philosophies of business model design prevail in the digital economy: "technology push"

Address correspondence to: Zhongfeng Su, Xi'an Jiaotong University, School of Management, No. 28, Xianning West Road, Xi'an, Shaanxi, 710049, P.R. China. E-mail: Zhongfengsu@163.com.

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and “market pull.” On the one hand, the diffusion of digital technologies such as artificial intelligence (AI), big data, and cloud computing is generating new business models, which aim to harness these sophisticated, cutting-edge technologies (Kohli and Melville, 2019; Sorescu, 2017; Teece, 2018). On the other hand, by enabling direct, frequent, and deep interactions between firms and their consumers, digital innovation is spawning numerous business models that can better identify, analyze, understand, and answer consumer needs (Priem et al., 2018; Wirtz, Schilke, and Ullrich, 2010).

We diverge from these existing viewpoints to argue that it is necessary to introduce a strategic orientation view of business model design in the context of the digital economy. In this study, we propose two strategic orientations for digital start-ups in business model design, namely, technology orientation and consumer orientation. Specifically, rooted in the resource-based view (RBV; Barney, 1991), technology orientation reflects the philosophy of “technology push,” which posits that consumers prefer technologically superior products or services (Gatignon and Xuereb, 1997). Conversely, based on the demand-side perspective (DSP; Priem, Li, and Carr, 2012), consumer

orientation, representing the philosophy of “market pull,” prioritizes the creation of superior consumer value (Narver and Slater, 1990). Consumer orientation is named so for two key reasons. First, market orientation is essentially customer orientation (Deshpandé, Farley, and Webster, 1993); second, we use the term “consumer” to incorporate customers and users, both of whom are important for firms operating in the digital economy (Abrell, Pihlajamaa, Kanto, Brocke, and Uebernickel, 2016). While both technology and consumer orientations have their merits, previous studies have shown inconsistent findings regarding their independent and interactive effects on firm performance (Hakala, 2011). Thus, we are curious about what strategic orientation a start-up should have when designing its business model, in order to stand out in an increasingly digital business environment.

We address this salient issue by drawing on both the RBV and the DSP to investigate strategic orientation’s role in successful business model design for digital start-ups. Specifically, this study aims to answer the following: (1) How do technology orientation and consumer orientation in business model design impact the performance of start-ups in the digital environment? (2) How does the borderless digital environment, featuring technological openness and user interactivity, moderate the relationship between strategic orientation and the performance of start-ups? We test the hypotheses with an on-site survey data set of digital start-ups. Empirical results show that both technology and consumer orientations are beneficial to the performance of digital start-ups. However, the pursuit of a balance between these two strategic orientations is harmful to firm performance. In addition, we find that the positive link between consumer orientation and firm performance is moderated by the digital business environment, in that it is strengthened by technological openness but weakened by user interactivity.

This study makes two contributions. First, it proposes and tests a strategic orientation view of business model design in the digital economy. This study adds value to existing literature by introducing two strategic orientations in business model design and investigating their impact on the performance of start-ups in the digital environment. We find that both technology and consumer orientations can improve the performance of start-ups independently. However, start-ups should make a choice between technology orientation and consumer orientation, instead of attempting to strike a balance between them. Second, this study contributes

BIOGRAPHICAL SKETCHES

Dr. Hai Guo is currently an associate professor at the School of Business, Renmin University of China. His research interests include entrepreneurship, technology and innovation management, and strategy in the context of digital economy. He has previously published his research in *Journal of Product Innovation Management*, *IEEE Transactions on Engineering Management*, *R&D Management*, *British Journal of Management*, *Journal of Business Research*, *Industrial Marketing Management*, *Management and Organization Review*, *Asia Pacific Journal of Management*, among others.

Ms. Chao Wang is currently a Ph.D. candidate at the School of Business, Renmin University of China. Her research interests are entrepreneurship, open innovation, and business model innovation.

Dr. Zhongfeng Su is a professor at the School of Management, Xi’an Jiaotong University. His research interests include strategic management, entrepreneurship, and innovation management. He has published over 50 papers in journals such as *Asia Pacific Journal of Management*, *International Journal of Production Research*, *Journal of Small Business Management*, *R&D Management*, *Marketing Letters*, *Management and Organization Review*, *British Journal of Management*, and *Small Business Economics*.

Dr. Donghan Wang is a professor at the School of Economics and Management, Communication University of China. His research interests are competitive strategy, technology, and innovation management. He has published in journals such as *IEEE Transactions on Engineering Management*, *Journal of Small Business Management*, *Small Business Economics*, *Journal of Knowledge Management*, among others.

to business model theory building by offering an integrative view of the RBV and the DSP. While technology orientation in business model design has its roots in the RBV, consumer orientation stems from the DSP. Our findings show that both the RBV and the DSP can serve to explain the performance implications of strategic orientation in business model design. Notably, our findings about the joint effect of technology orientation and consumer orientation indicate that the two theories are substitutable, but findings about the contingent value of digital business environments provide evidence for their complementarity.

Theory and Hypotheses

Defining the Business Model

What is a business model? Although the term “business model” has garnered plenty of attention in both theory and practice (Foss and Saebi, 2017), the concept is controversial (Massa, Tucci, and Afuah, 2017). Positioning itself in the strategy field (Wirtz, Pistoia, Ullrich, and Göttel, 2016), this study integrates two influential views to define a business model. On the one hand, the business logic view describes a business model as “a representation of a firm’s underlying core logic” (Shafer, Smith, and Linder, 2005, p. 202), which guides strategy execution in order to create value (Casadesus-Masanell and Ricart, 2010; Richardson, 2008). On the other hand, the architectural view considers the business model to be a multi-dimensional construct that defines the holistic architecture of the business, comprising three central elements: value proposition, value creation, and value capture (e.g., Baden-Fuller and Haefliger, 2013; Teece, 2010). This study combines the above to define a *business model* as the core business logic that an organization employs to create, deliver, and capture value (Teece, 2010).

How does a business model create value? Fragmented theoretical explanations exist in the literature (Amit and Zott, 2001; Dasilva and Trkman, 2014; Teece, 2010). For a long time, the business model’s value was explained by various supply side theories, such as the RBV (DaSilva and Trkman, 2014; Zott and Amit, 2010), the capability-based view (Mezger, 2014), and the dynamic capabilities perspective (Teece, 2018). Scholars now increasingly recognize the importance of customers (consumers) in business model theory development (Keininghama et al., 2019; Rietveld, 2018), leading to a demand-side explanation (Priem et al., 2018). In this study, we integrate the RBV and

the DSP to explain the role of the business model for new venture growth. The RBV argues that the possession of heterogeneous resources can give a focal firm a competitive advantage (Barney, 1991). According to the RBV, the business model is key to help start-ups mobilize resources and exploit opportunities (e.g., Amit and Han, 2017; Teece, 2018; Zott and Amit, 2010). For example, Chesbrough and Rosenbloom (2002) consider the business model as an intermediary to convert technological resources into economic value (Baden-Fuller and Haefliger, 2013; Dmitriev, Geoff, Truong, Palmer, and Schneckenberg, 2014). Meanwhile, shifting attention from the focal firm toward product markets and consumers, the DSP argues that the heterogeneity of consumers and the dynamic changes of consumer preferences largely determine a firm’s managerial decisions about value creation and consequently lead to performance differences (Adner and Snow, 2010; Priem et al., 2012). According to the DSP, the business model articulates how an organization creates value by looking downstream toward product markets and consumers (Priem et al., 2018; Rietveld, 2018).

Defining the Digital Business Environment

Compared to the traditional business environment, the new digital world is less bounded and predefined (Nambisan, 2017). Regarding the technological dimension, a less bounded digital environment is characterized by a highly open technological environment. As noted by Richter, Kraus, Brem, Durst, and Giselbrecht (2017, p. 300), the sharing economy where “an increasing number of people are participating in sharing and exchanging information, knowledge, data” has come into effect in the digital age. New concepts or ideas such as open innovation, intellectual commons, peer production, and collective invention prevail in the sharing economy. In such an environment, firms could hardly build and maintain a permanent technological barrier (Chesbrough and Appleyard, 2007). Instead, they turn to making or using open source technologies by adopting an open strategy (Hautz, Seidl, and Whittington, 2017). In this study, we define *technological openness* as an environment where technology-related information, knowledge, and resources can be freely accessed, exchanged, or shared with low costs.

Regarding the market dimension, the less bounded digital environment features high-level interactions

between firms and their consumers. According to Prahalad and Ramaswamy (2000), *user interactivity* refers to an environment where consumers are no longer onlookers, but instead frequently participate in active and explicit dialogues with suppliers through various channels for better products and services. The development of digital technologies, platforms, and infrastructure offers consumers abundant opportunities to learn about businesses by communicating directly with suppliers or other consumers (Nambisan, 2002; Priem et al., 2018). Furthermore, these interactive dialogues are often initiated by consumers, no longer being controlled by the firm (Prahalad and Ramaswamy, 2000). For example, consumers have transformed into co-creators or “prosumers” in the process of new product development (Hiennerth, Lettl, and Keinz, 2013).

Strategic Orientation in Business Model Design

How should a start-up design its business model? Existing literature offers two useful perspectives. Regarding business model design as a process of experimentation, learning, and adaptation (Sosna et al., 2010), *the evolutionary perspective* addresses issues such as how entrepreneurs effectively design business models in a nascent market (McDonald and Eisenhardt, 2020), how business model innovation emerges in new ventures (Snihur and Zott, 2020), how established firms transform or innovate their business models (Aspara et al., 2013; Berends et al., 2016), and how the business model evolves in order to adapt to the changing environment (Demil and Lecocq, 2010). Emphasizing the key thought in a business model design, *the thematic-based perspective* argues that the business model can be characterized by design themes that detail the business model’s dominant value creation sources (Zott and Amit, 2010). The most commonly used design themes are called NICE: *novelty*, *lock-in*, *complementarity*, and *efficiency* (Amit and Zott, 2001; Kulins et al., 2016; Pati et al., 2018; Zott and Amit, 2007, 2008, 2010).

The new digital environment provides a fresh context for business model research (Amit and Han, 2017; Nambisan, 2017). The emergence of novel digital technologies not only generates abundant commercialization opportunities (Kohli and Melville, 2019; Sorescu, 2017; Teece, 2018), but also enables start-ups to better understand consumer needs by building close and interactive relationships with consumers

(Priem et al., 2018; Shah and Tripsas, 2007; Wirtz et al., 2010). Thus, advanced technologies and consumer needs become two key driving forces for value creation by digital start-ups.

Departing from existing literature, the present study endeavors to propose a strategic orientation view of business model design. Reflecting the firm’s philosophy of how to conduct business through a set of values and beliefs (Gatignon and Xuereb, 1997), strategic orientation is important in guiding the firm to allocate resources and undertake activities in order to achieve superior performance (Zhou, Yin, and Tse, 2005). In this sense, strategic orientation is particularly valuable for business model design, because the business model is embodied as an activity system that guides strategy execution (Casadesus-Masanell and Ricart, 2010; Richardson, 2008; Zott and Amit, 2010). Drawing on insights from the strategic orientation literature (Deshpandé et al., 1993; Gatignon and Xuereb, 1997; Hakala, 2011; Zhou et al., 2005), we propose two strategic orientations in the area of business model design, namely, technology orientation and consumer orientation.

Reflecting the “technology push” philosophy, *technology orientation* refers to a start-up’s inclination to introduce or use new technologies, products, or innovations in business model design. In this case, investment in technologies is prioritized and value is created and delivered through developing and adapting new technologies (Gatignon and Xuereb, 1997). According to the RBV (Barney, 1991), technological resources are a key source of firm innovation and core competence development (Hitt, Bierman, Shimizu, and Kochhar, 2001). Start-ups that adopt technology-oriented business models will first look upstream toward sources of advanced digital technologies, and then, find ways to turn technologies into economic value (Ross, Sebastian, and Beath, 2017). Meanwhile, rooted in the “market pull” philosophy, *consumer orientation* refers to a start-up’s tendency to gain sufficient understanding of the target consumers and find ways to deliver superior value to them through the business model design (Narver and Slater, 1990). In this case, consumer needs serve as the primary source of value creation, preceding a start-up’s likely investment in technologies. The DSP insists that “resources gain economic value from their use by customers” (Kor, Mahoney, and Michael, 2007, p. 1198). Thus, the essence of consumer orientation is to “look downstream from the focal firm, toward product markets and consumers” (Priem et al., 2012, p. 346), then find

the solution to satisfy consumer needs, and benefit from it (Adner and Snow, 2010).

Strategic Orientation in Business Model Design and the Performance of Digital Start-ups

According to the RBV, technology orientation in business model design can lead to improved performance for digital start-ups. First, a technology-oriented business model may generate entrepreneurial rents by leveraging advanced technological resources. These economic rents may accrue to all business model stakeholders between the introduction of a new technology (or new technological allocation) and its diffusion. That is, by drawing on new technologies, a technology-oriented business model can bridge the factor and product markets in novel or more efficient ways (Johnson, Friend, and Lee, 2017; Kroh, Luetjen, Globocnik, and Schultz, 2018). Second, a technology-oriented business model may in itself be part of the new market development process. Given the bounded rationality of consumers, partly shaped by the technological environment (Priem et al., 2012), many potential opportunities cannot be recognized (Hamel and Prahalad, 1991). Business models centered on disruptive technologies cater to their latent needs (Beckman, Eisenhardt, Kotha, Meyer, and Rajagopalan, 2012; Teece, 2010). For example, based on its strong computing power, Google successfully created the search engine market. Google has recently invested heavily in the area of self-driving vehicles, a new market primarily driven by AI technologies. Thus, we suggest the following:

H1: Technology orientation in business model design is positively related to digital start-up performance.

According to the DSP, consumer orientation in business model design can also improve the performance of digital start-ups. First, a consumer-oriented business model may enhance economic efficiency, by reducing costs or risks of resource investment, as well as enhancing consumers' willingness to pay (Baden-Fuller and Haeffliger, 2013). While the business model is anchored on a focal firm, it is market-centric so as to enable the focal firm to appropriate a share of the greater value created (Zott and Amit, 2013). A start-up with a better understanding of consumer needs can maximize value capture efficiency via accurate resource investment. Second, a consumer-oriented business model enables demand-related information to flow better into

the focal firm, contributing to the recognition of new market opportunities. Consumers are sometimes the first to perceive the potential value of new technologies (Shah and Tripsas, 2007; Teece, 2010) and participate in value creation (Visnjic, Wiengarten, and Andy, 2016). Xiaomi, the well-known Chinese internet company, built the virtual "millet community" ("xiaomi" is Chinese for "millet") where its fans are encouraged to discuss their needs. With a deep understanding of China's new middle class and relying on its unique capabilities in user relationship management, Xiaomi is able to introduce a variety of new products and services to the market. Thus, we propose the following:

H2: Consumer orientation in business model design is positively related to digital start-up performance.

According to the literature review of Hakala (2011), there are three perspectives explaining the joint effect of multiple strategic orientations: sequential, alternative, and complementary. Meanwhile, scholars are calling for the integration of the RBV and the DSP in strategy and entrepreneurship research (Amit and Han, 2017; Esslinger, 2011; Priem, Butler, and Li, 2013; Ritter and Lettl, 2018). In this study, we argue for an alternative perspective, that while both technology and consumer orientations in business model design are beneficial, digital start-ups should take one dominant strategic orientation in business model design.

First, technology orientation and consumer orientation reflect two competing business philosophies in the digital economy. Based on the RBV, the logic of the "technology-push" philosophy is that a firm's idiosyncratic and difficult-to-imitate technological resources enable it to achieve greater performance (Zhou et al., 2005). Thus, technology orientation emphasizes a firm's internal technology capabilities as the starting point for business model design. Therefore, the start-up employs start-of-the-art technologies to exploit opportunities. Based on the DSP, the logic of the "market-pull" philosophy is that for a firm to achieve superior performance, it must create value for the customer (Narver and Slater, 1990). Thus, consumer orientation emphasizes a firm's external market capabilities as the starting point for business model design. Therefore, the start-up exploits opportunities by identifying, analyzing, understanding, and answering the consumer's expressed needs (Gatignon and Xuereb, 1997). We argue, however, that the transformational or disruptive nature of digital innovation requires digital start-ups to hold either

a “technology-push” or a “market-pull” philosophy in business model design. Start-ups that choose a technology-oriented business model cannot over-emphasize consumers, and should even ignore them, because they are not knowledgeable about the latest technologies and the market trends in times of disruptive changes (Teece, 2018; Zhou et al., 2005). Meanwhile, start-ups that choose a consumer-oriented business model should not emphasize seeking technological breakthroughs because consumers are often shortsighted and care more about how to meet their expressed needs, instead of needs they are unaware of (Narver, Slater, and McLachlan, 2004; Zhou et al., 2005).

Second, technology orientation and consumer orientation would compete for a digital start-up’s scarce resources. Given the limited resources available to start-ups (De Massis, Audretsch, Uhlaner, and Kammerlander, 2017), attempts to embrace both strategic orientations simultaneously could lead to suboptimal resource allocation. The resource constraint may result in a lack of focus, which may confuse market participants, create technological and organizational problems, and lead to high costs (Zott and Amit, 2007). A firm that stalemates between the resource side and the market side—or, analogously, between technology orientation and consumer orientation—may perform poorly because it misses out on the opportunity to learn to become an even more unique “technician” or “market hunter” (Ross et al., 2017). Therefore, we believe that it is better for a digital start-up to concentrate on one of the two design orientations rather than to incorporate both simultaneously, and hypothesize the following:

H3: Balancing technology orientation and consumer orientation in business model design is negatively related to the performance of start-ups in the digital environment. In other words, leaning heavily to either technology orientation or consumer orientation is positively related to digital start-up performance.

The Contingent Value of Technological Openness and User Interactivity

The impact of strategic orientation on firm performance is contingent to the external environment (e.g., Gatignon and Xuereb, 1997; Narver and Slater, 1990; Zhou and Li, 2007). In this study, we suggest that the

digital environment’s borderless nature, manifested particularly by technological openness and user interactivity, will moderate the relationship between strategic orientation in business model design and the performance of start-ups.

We argue that technology orientation and technological openness are substitutive in contributing to the performance of digital start-ups. In an open technological environment, start-ups can easily access external technological resources such as open source technologies or technical talent. According to the RBV, these widely available resources could hardly become sources of a firm’s competitive advantage (Barney, Wright, and Ketchen, 2001), making it difficult for a start-up to “build a moat” by investing in, commercializing, and benefiting from technologies (Chesbrough and Appleyard, 2007). In other words, technological openness will lead to a highly munificent technological environment and make a start-up’s investment in technologies highly risky.

In contrast, we suggest that technology orientation and user interactivity can complement each other to foster the growth of digital start-ups. Start-ups that adopt technology-oriented business models tend to invest in new technologies and create value through new technological solutions (Gatignon and Xuereb, 1997). According to the DSP, a technology’s economic value depends on whether it can satisfy consumer needs (Priem et al., 2012). Operating in a highly user interactive environment, start-ups can provide consumers with opportunities to better understand the technologies as well as their business potential. Therefore, they can awaken their consumers at a low cost and address the consumers’ latent needs through unique technological solutions (Narver et al., 2004). Therefore, we propose the following:

H4: The relationship between technology orientation in business model design and digital start-up performance is negatively moderated by technological openness.

H5: The relationship between technology orientation in business model design and digital start-up performance is positively moderated by user interactivity.

We suggest that consumer orientation and technological openness are complementary in improving digital start-ups’ performance. As mentioned earlier, consumer orientation guides value creation by

identifying, analyzing, understanding, and answering consumer needs (Gatignon and Xuereb, 1997). According to the RBV, the process of value creation and delivery is inseparable from the support of resources, providing solutions to meet consumer needs (Barney, 1991). Without the ability to create and deliver value, even perfect value proposition cannot be achieved (Ross et al., 2017). Fortunately, an open technological environment offers opportunities for consumer-oriented business models to easily create value. Since technological resources such as new technologies are readily available or imitable from commercial sources, start-ups are able to quickly learn or imitate them at lower “copy” costs (Richter et al., 2017).

Meanwhile, we argue that consumer orientation and user interactivity are substitutive in contributing to the performance of digital start-ups. In order to tap into consumer needs, consumer-oriented business model design requires a start-up to invest heavily in activities such as market analysis and consumer relationship management. However, in environments where consumers can interact with firms frequently, extensively, and instantly, consumer demands can be easily perceived and captured at relatively low costs (Priem et al., 2018). For example, start-ups can easily get user feedback through channels such as virtual communities, apps, and social networking services (Priem et al., 2018), with the help of data analysis tools (Richter et al., 2017; Sorescu, 2017). Therefore, in highly user interactive environments, the importance of the start-up’s investment on the demand side will be diminished. In addition, excessive emphasis on the demand side might hinder a start-up’s development of internal technology capabilities (Demil and Lecocq, 2010). Thus, we suggest the following:

H6: The relationship between consumer orientation in business model design and digital start-up performance is positively moderated by technological openness.

H7: The relationship between consumer orientation in business model design and digital start-up performance is negatively moderated by user interactivity.

Methodology

Sample and Data

To test our hypotheses, we employed the questionnaire survey method and developed a dataset of

Chinese digital start-ups. China is an ideal empirical setting for this study for several reasons. First, China has become one of the largest and most active digital economies in the world. In 2018, the size of China’s digital economy grew to 31.3 trillion RMB, accounting for 34.8% of the GDP. Second, China is paying special attention to digital entrepreneurship. In 2014, the Chinese central government released the “mass entrepreneurship and innovation” initiative to promote technological innovation, stimulate industrial upgrading, and build a new economy. According to a discussion paper from the McKinsey Global Institute released in 2017, China was among the top three countries for venture-capital investment in key digital technologies. Finally, novel digital technologies and business models constantly emerge in China, making both technology- and consumer-oriented business models flourish and leading to an increasingly digital business environment.

Our survey was conducted in the summer of 2017, by three research teams comprising doctoral students. We organized two meetings to train the investigators to conduct questionnaire surveys. We chose on-site surveys over online- or mail-based surveys to minimize potential misinterpretations of items and improve the response rate. The process typically flowed thus: First, an interviewer contacted the interviewee to ask for permission and make an appointment. Second, during the on-site survey, the interviewer interpreted the purpose of the study and gave instructions on how to fill in the questionnaire. Finally, the interviewer double-checked the completed questionnaire.

We collected data from five provinces or municipalities of China: Beijing, Zhejiang, Shandong, Hunan, and Anhui. Beijing and Hangzhou represent two typical cities for digital entrepreneurship. Known as China’s Silicon Valley, the Zhongguan Village located in Beijing is the center for technology entrepreneurship. In Zhejiang province, the city of Hangzhou is famous for its e-business and internet economy. With the help of local governments (e.g., the Management Committee of Zhongguancun), we drew a random sample of 500 start-ups located in 34 incubators or pioneer parks. By the end of October 2017, 389 firms joined the survey and filled out our questionnaires, yielding a 77.8% response rate. Among them, 108 were deleted due to three reasons. First, the questionnaire was filled in less than 10 minutes, an indication of low quality. Second, the questionnaire was not completely filled.

Third, firms were older than 10 years at the time of data collection (Nason, Wiklund, Mckelvie, Hitt, and Yu, 2019). Thus, the final sample contains 281 firms, leading to an effective response rate of 56.2% (281/500).

Measures

Dependent variables. Scholars insist that it is inappropriate to measure the performance of start-ups using traditional accounting-based indicators (Shane and Stuart, 2002; Wang, Thornhill, and De Castro, 2017). Rather, the success of a start-up is better reflected by its growth potential (Brinckmann, Salomo, and Gemuenden, 2011; Gilbert, McDougall, and Audretsch, 2006). In this study, we used *sales growth* rate and *user growth* rate to measure the performance of digital start-ups, respectively. Both were self-reported objective indicators.

Independent variables. A systematic review of business model literature revealed that only a few quantitative measurement scales for business model design exist (e.g., Zott and Amit, 2007), none of which examine strategic orientation. For the purpose of this study, we developed a new measurement scale for *strategic orientation in business model design* by following the multi-step approach of Churchill (1979) and Schrauder, Kock, Baccarella, and Voigt (2018).

First, we specified the domain of the construct and defined the business model as the underlying business logic that an organization employs to create, deliver, and capture value (Teece, 2010). We identified key items related to the strategic orientation of value creation, value proposition, and value capture when start-ups design their business models. This step was performed by consulting the existing literature on business models and in discussions and workshops with scholars and practitioners. Referring to existing studies (Clauss, 2017; Schrauder et al., 2018; Teece, 2010), we theoretically specified strategic orientation in business model design as a second-order construct consisting of three sub-constructs of value creation design, value proposition design, and value capture design.

Second, we generated a sample set of items. This step was based on the literature review on existing measures from both business model (design) research (Clauss, 2017; Schrauder et al., 2018; Teece, 2010;

Zott and Amit, 2007) and strategic orientation research (Brown, Davidsson, and Wiklund, 2001; Ross et al., 2017), as well as on multiple workshops with experienced academics and practitioners.

Drawing from the scale for strategic orientation (Brown et al., 2001; Ross et al., 2017), the items were treated as forced-choice types, with pairs of statements representing the opposite ends of the consumer-orientation/technology-orientation continuum. In order to avoid response set contamination, that is, the tendency to respond to statements for reasons other than their content, the questions were arranged so that consumer-orientation and technology orientation statements appeared on both the left and right sides (Brown et al., 2001). A 5-point scale divided the two statements.

We organized three rounds of interviews to polish the scale. In the first round, we asked for comments from three experienced professors, in order to ensure all scales and items were rooted in the literature. In the second round, we asked for comments from three experienced entrepreneurs, in order to ensure that our questions reflected business practices. In the third round, we further asked for comments from three digital start-up founders, ensuring all scales and items were accurate and could be easily understood. In the final round, in order to ensure the questionnaire's content and construct validity, we organized a pilot study among ten founders (or co-founders) of digital start-ups, based on which the scale was completed and applied.

Finally, to assess the factor structure, we specified and tested the measurement model in SmartPLS3.0. According to previous studies (Clauss, 2017; Schrauder et al., 2018; Teece, 2010), strategic orientation in business model design was regarded as a formative construct. Specifically, we specified the construct as a type IV formative-formative model (Becker, Klein, and Wetzels, 2012; Jarvis, MacKenzie, and Podsakoff, 2003; Wetzels, Odekerken-Schroder, and Van Oppen, 2009). Following the recommendations of Becker et al. (2012), we applied the repeated indicator approach to specify the hierarchical structure in PLS. For this approach, a higher order latent variable was represented by underlying first-order latent variables and all manifest variables of the underlying lower order constructs. Therefore, the items were used twice in the case of second-order models.

Traditional tests for indicator reliability and convergence are not necessary for formative models, as there

is no a priori assumption about the strength of the relationship between first-order and second-order latent variables (Hulland, 1999). In contrast, multicollinearity is an issue specific to formative indicators. This is because “the formative measurement model is based on a multiple regression, and, therefore, the stability of the indicator coefficients is affected by the sample size and strength of the indicator intercorrelations” (Diamantopoulos and Winklhofer, 2001, p. 272). Excessive collinearity among indicators thus makes it difficult to separate the distinct influence of the individual indicator on the latent variable. Therefore, we calculated the variance inflation factors of the lower order constructs. As shown in Appendix A, scores all ranged between 1.271 and 1.840, significantly below the threshold value of 10.

For formative indicators, which have a regression-like relationship with the latent construct, only the weights (and not the loadings) need to be considered in assessing the measurement model (Klein and Rai, 2009). While no minimum threshold values for indicator weights have been established, the statistical significance of the weights can be used to determine the relative importance of the indicators in forming a latent variable (Ravichandran and Rai, 2000). Figure 1 shows that all first- and second-order weights are significant. Therefore, the hierarchical measurement model provides an appropriate measurement approach.

Conventional procedures used to assess the validity and reliability of scales composed of reflective

indicators (e.g., factor analysis and assessment of internal consistency) are not appropriate for formative indicators (Jarvis et al., 2003). “The best we can do ... is to examine how well the index relates to measures of other variables” (Diamantopoulos and Winklhofer, 2001, p. 272). We, therefore, tested the predictive power of the structural model on two additional variables (i.e., sales growth and user growth) in PLS. Based on our hypotheses, we tested the U-shaped relationship between the second-order measure of strategic orientation in business model design and the two constructs. Strategic orientation in business model design explained 5.8% of the variance of sales growth ($R^2 = .058$) and 3.7% of the variance of sales growth ($R^2 = .037$). As assumed, the square of strategic orientation in business model design has significant effects on sales growth ($\beta = .194, p < .001$) and user growth ($\beta = .152, p < .05$), which, therefore, confirm the nomological validity of the scale.

Moderating variables. We developed two scales for moderating variables as well. As previously mentioned, scholars have increasingly attempted to depict the digital environment, from both the demand side and technology side (e.g., Nambisan, 2017; Priem et al., 2018). First, based on existing literature, we specified the domain of the constructs and defined *technological openness* and *user interactivity* as previously mentioned. Second, we generated a sample set of items. This step was based on existing measures and concepts from digital entrepreneurship,

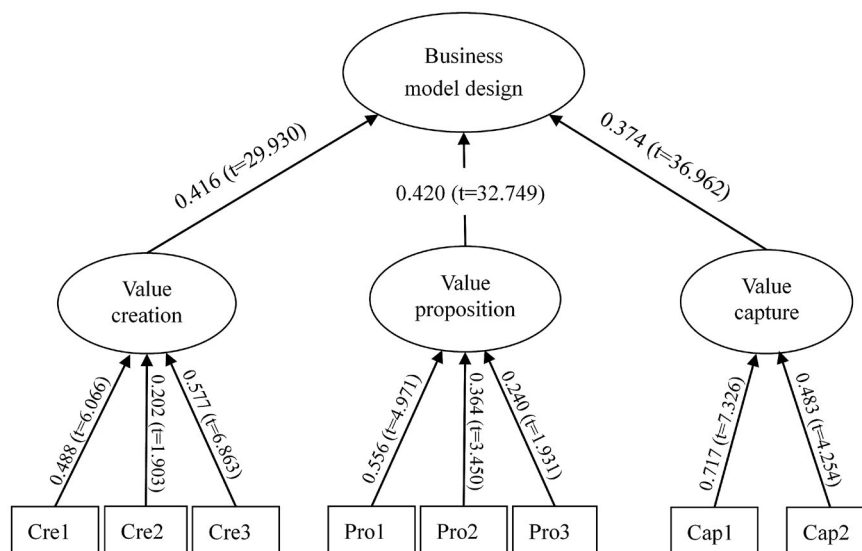


Figure 1. Hierarchical Measurement Model of Strategic Orientation in Business Model Design

digital innovation, and organizational environment research (Beckman et al., 2012; Hienerth et al., 2013; Nambisan, 2002, 2017; Prahalad and Ramaswamy, 2000; Priem et al., 2018; Richter et al., 2017), as well as on multiple workshops with academics and practitioners. Similarly, we organized three rounds of interviews to polish the scale.

Finally, we used the quantitative feedback to refine the measures by checking their reliability and validity. Based on an exploratory factor analysis, the final technological openness scale consists of five items (one item with low loading was removed) and user interactivity scale consists of three items. All the items have high loadings (i.e., greater than .7) on their respective constructs. We then applied confirmatory factor analysis to validate the scales. As shown in Appendix B, the reliability and validity of the scale were acceptable because all factor loadings were well above .7 and most α values were greater than .6. In addition, we tested validity of these constructs, and the model fit is reasonably good ($\chi^2 = 77.000$, $df = 19$, $\chi^2/df = 4.053$, GFI = .933, NFI = .911, CFI = .931).

Control variables. Twelve variables were controlled for this study. Three individual-level factors were controlled: *founder age*, *founder start-up experience*, and *founder learning experience*. Founder age was divided into five classes, with 1 indicating younger than 20 years old, 2 indicating 21–30, 3 indicating 31–40, 4 indicating 41–50, and 5 indicating older than 50. Start-up experience was treated as a dummy variable, with 1 indicating the respondent had started a business before, and 0 otherwise. Similarly, learning experience was also treated as a dummy variable, with 1 indicating the respondent had at least one of the following experiences: studied in a business school, participated in entrepreneurial training programs, or studied abroad for at least 1 year, and 0 otherwise.

We controlled six firm- or team-level variables. Following Zott and Amit (2007), we controlled *firm age* and *firm size* (measured by the number of full-time employees). We measured *founding team size* as the number of founding team members, because it may imply different levels of access to resources and team heterogeneity, in turn leading to different viewpoints about business model design and distinct performance outcomes (Muñoz, Sanchez-Bueno, and Vos-Saz, 2015). In line with existing literature, business model design is highly related to the organization

design (Amit and Zott, 2001). Hence, we controlled *R&D team size* (number of technological personnel) and *marketing team size* (number of marketing personnel). Considering the impact of investors on the strategic behavior and the performance of start-ups (Huang and Knight, 2017), we measured *investor involvement* as a dummy variable, with 1 indicating the start-up has been funded by at least one venture capital (VC) and 0 otherwise.

Finally, we controlled three industrial- or environmental-level variables. We measured *industry* according to the industry classification of China Securities Regulatory Commission. Following Child (1972) and Schrauder et al. (2018), we controlled for *institutional uncertainty*, measured by the self-reported “degree of changes of relevant policies in the last three years,” and *technological turbulence*, measured by “the degree of technological changes within your field in the last three years.”

Addressing Common Method Variance

We used two methods to control for common method variance (CMV). First, we used two informants rather than one single informant from each start-up for data collection (Zhang and Li, 2010). We divided our scales into two questionnaires, namely Part A and Part B. Questions for Part A of the questionnaire were related to business model design, which were filled out by one founder (co-founder) of the start-up. Questions for Part B were related to the external environment and firm performance, which were filled out by another founder (co-founder). Second, we used Harman’s one-factor test to check for the severity of CMV. Our results showed that the first extracted factor explained only 15.90% of the variance, indicating that no single factor would affect the studied constructs simultaneously. Thus, CMV is unlikely to be a serious problem of this study.

Results

Hypotheses Testing

Table 1 depicts the means, standard deviations, and the Pearson correlations among the variables. The correlations among the explanatory variables are relatively low, with a maximum absolute value of .156. In order to avoid the possible threat of multicollinearity, we zero-centered all variables before regression.

Table 1. Descriptive Statistics and Correlation Matrix

	Mean	S.D.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1. Sales growth	.765	1.397	1														
2. User growth	1.067	1.724	.823***	1													
3. SOBMD	.000	1.000	-.018	.041	1												
4. Technological openness	3.473	.844	.056	.020	-.021	1											
5. User interactivity	4.084	.784	-.141*	-.166**	-.110	.156***	1										
6. Firm age	2.916	2.124	-.045	-.067	.038	.066	.006	1									
7. Firm size	32.120	81.896	-.016	-.001	-.021	.071	.058	.163***	1								
8. Founding team size	5.313	2.663	-.012	.017	.116	.057	-.104	.051	.133*	1							
9. R&D team size	10.121	15.912	-.045	-.017	.093	.082	-.016	.283***	.687***	.296***	1						
10. Marketing team size	6.065	7.682	.000	-.010	-.079	.129*	-.020	-.035	.016	.026	.078	1					
11. Investor involvement	.263	.429	-.001	.083	-.031	.018	-.036	-.080	.182**	.062	.172**	-.020	1				
12. Founder age	2.707	.708	.138*	.091	.030	.060	-.087	.284***	.048	-.032	.146*	-.061	-.060	1			
13. Start-up experience	.324	.469	-.003	-.017	-.129*	.099	.023	-.140*	-.042	.010	.046	.133*	.091	.225***	1		
14. Learning experience	.541	.499	.064	.103	.014	.091	.023	.021	-.018	.126*	.060	-.053	.073	.008	.149*	1	
15. Institutional uncertainty	3.457	.948	.003	.014	.025	.022	-.006	.012	.057	-.084	.083	.017	.062	.007	.140*	.015	1
16. Technological turbulence	3.612	1.072	.016	.059	.031	.383***	.058	.053	.099	.007	.077	.004	.157**	.040	.060	.073	.132*

Note: SOBMD = strategic orientation in business model design; *, **, and *** denote significance at the 5%, 1%, and .1% levels.

To test our hypotheses, we divided our sample into two groups with the help of K-means cluster analysis (Bigliardi, Nosella, and Verbano, 2005). The grouping variables are the three first-order factors of strategic orientation in business model design. As shown in Table 2, the two groups are quite different, with Group 2 (117 firms) showing significantly higher values than Group 1 (164 firms) in all three dimensions. Thus, Group 1 was classified as the consumer-oriented sample, and Group 2 as the technology-oriented sample. This result shows that more digital start-ups design consumer-oriented business models in China. In order to assess the validity of our clustering results, we further conducted ANOVA analysis (Bigliardi et al, 2005). Results in Table 2 clearly show that all factors between the two clusters are statistically different.

Table 3 reports a series of analyses testing the hypotheses. All hypotheses regarding the main effects of strategic orientation in business model design (H1–H3) are supported. Specifically, Models 1, 4, and 7 include controls only. Model 2 shows that for start-ups whose business models are technology-oriented, business model design is positively related to sales growth ($\beta = .587, p < .01$), which is consistent with H1. Model 5 shows that for start-ups whose business models are consumer-oriented, business model design is negatively related to sales growth ($\beta = -.683, p < .01$). It is noteworthy that the coefficient for consumer-oriented business model design should be reversely interpreted, because it lies at the low end of the continuum. Thus, this result suggests that the more a start-up's business model is consumer-oriented, the higher the sales growth, thus providing support for H2. Model 8 and Model 9 are results for the full sample. Model 8 indicates that there is no linear relationship between strategic orientation in business model design and the performance of start-ups ($\beta = -.056, p = .533$). However, Model 9 shows that the squared term of strategic orientation in business model design is positively

related to sales growth ($\beta = .313, p < .001$), indicating a U-shaped relationship. This result provides support for H1, H2, and H3 simultaneously, by suggesting that while both technology- and consumer-oriented business model design are beneficial for the performance of start-ups, the pursuit of a balance between the two strategic orientations is counterproductive. Next, we replaced sales growth with user growth and reran all the regressions. As shown in Table 4, testing results for H1, H2, and H3 are highly consistent.

Models 3 and 6 in Table 3 test the moderating effects of technological openness and user interactivity. The contingency hypotheses regarding consumer-oriented business model design (H6 and H7) are supported, yet those regarding technology-oriented business model design (H4 and H5) fail to be supported. As shown in Model 3, for start-ups that design technology-oriented business models, the moderating effects of technological openness ($\beta = .082, p = .724$) and user interactivity ($\beta = -.037, p = .870$) are both insignificant. Thus, H4 and H5 are not supported by our data. Model 6 shows that for start-ups that design consumer-oriented business models, technological openness positively moderates the relationship between business model design and sales growth ($\beta = -.362$, reversed scoring, $p < .1$), but user interactivity negatively moderates this relationship ($\beta = 1.293$, reversed scoring, $p < .001$). These results provide support for H6 and H7. Similarly, we replaced sales growth with user growth and reran all the regressions. As shown in Table 4, testing results for hypotheses regarding the moderating effects are also consistent.

To facilitate interpretation, we plotted the effect of strategic orientation in business model design on the performance of start-ups. As shown in Figure 2, when the level of either consumer orientation or technology orientation is higher, the sales growth rate of digital start-ups becomes higher. However, a moderate level between consumer orientation and technology orientation leads to lower sales growth.

Table 2. K-means Cluster Analysis and ANOVA Analysis

	Group 1 (mean)	Group 2 (mean)	Cluster		Error		<i>F</i>
			MS between groups	df	MS within groups	df	
Value creation	-.610	.850	146.247	1	.483	279	302.807***
Value proposition	-.640	.900	161.543	1	.428	279	377.236***
Value capture	-.410	.570	66.089	1	.770	279	85.792***
<i>N</i>	164	117	–	–	–	–	–

Note: *** denotes significance at the .1% levels.

Table 3. Results of the OLS Regression Analysis for Sales Growth

	Technology-oriented sample			Consumer-oriented sample			Full sample		
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8	Model 9
Firm age	.063 (.059)	.056 (.056)	.058 (.057)	-.126 [†] (.064)	-.123 [†] (.063)	-.119* (.054)	-.044 (.045)	-.045 (.045)	-.044 (.043)
Firm size	.000 (.004)	.001 (.004)	.001 (.004)	.001 (.002)	.000 (.002)	.000 (.002)	.001 (.001)	.001 (.001)	.001 (.001)
Founding team size	-.012 (.035)	-.017 (.033)	-.017 (.034)	.036 (.078)	.025 (.076)	.054 (.065)	-.007 (.034)	-.006 (.034)	-.008 (.033)
R&D team size	-.012 (.010)	-.010 (.010)	-.010 (.010)	-.004 (.015)	.005 (.015)	.007 (.013)	-.009 (.008)	-.009 (.008)	-.006 (.008)
Marketing team size	-.010 (.025)	-.004 (.024)	-.005 (.024)	.004 (.014)	-.003 (.014)	-.019 (.012)	.005 (.011)	.004 (.011)	.001 (.011)
Investor involvement	-.016 (.307)	-.056 (.293)	-.053 (.303)	-.327 (.304)	-.277 (.297)	-.358 (.256)	-.036 (.209)	-.042 (.210)	-.040 (.202)
Founder age	.084 (.176)	.036 (.168)	.032 (.171)	.451* (.193)	.514** (.190)	.365* (.163)	.308* (.133)	.309* (.133)	.300* (.128)
Start-up experience	.258 (.269)	.369 (.259)	.376 (.267)	-.442 (.298)	-.675* (.303)	-.675** (.259)	-.163 (.199)	-.177 (.200)	-.223 (.194)
Learning experience	-.038 (.231)	-.072 (.220)	-.086 (.225)	.349 (.251)	.302 (.246)	.272 (.210)	.225 (.173)	.226 (.173)	.168 (.168)
Institutional uncertainty	-.092 (.127)	-.043 (.122)	-.044 (.124)	.040 (.137)	.051 (.134)	.126 (.115)	.017 (.093)	.018 (.093)	.052 (.090)
Technological turbulence	.296* (.134)	.286* (.127)	.286* (.133)	-.119 (.118)	-.103 (.115)	-.054 (.100)	.002 (.088)	.003 (.088)	.013 (.085)
Technological openness	-.039 (.145)	-.084 (.139)	-.160 (.270)	.148 (.134)	.140 (.131)	-.047 (.176)	.089 (.096)	.090 (.096)	.070 (.093)
User interactivity	.080 (.129)	-.001 (.125)	.033 (.241)	-.307* (.122)	-.377** (.122)	.527** (.170)	-.192* (.089)	-.197* (.090)	-.266** (.088)
Industry	Control	Control	Control	Control	Control	Control	Control	Control	Control
SOBMD		.587** (.181)	.573** (.186)		-.683** (.245)	-1.061*** (.220)		-.056 (.090)	-.146 (.089)
SOBMD ²									.313*** (.071)
SOBMD × Technological openness			.082 (.231)			-.362 [†] (.215)			
SOBMD × User interactivity			-.037 (.228)			1.293*** (.187)			
Constant	-.276 (.866)	-.902 (.847)	-.871 (.876)	-.319 (1.753)	-.559 (1.713)	1.017 (1.483)	-.295 (1.507)	-.303 (1.509)	-.419 (1.457)
R ²	.163	.250	.251	.249	.290	.490	.116	.117	.181
Adjusted R ²	-.055	.044	.024	.093	.137	.370	.018	.015	.083
F value	.748	1.213	1.106	1.600*	1.890**	4.089***	1.180	1.150	1.847**

Note: SOBMD = strategic orientation in business model design; Standard errors are reported in the parentheses; †, *, **, and *** denote significance at the 10%, 5%, 1%, and .1% levels.

Table 4. Results of the OLS Regression Analysis for User Growth

	Technology-oriented sample			Consumer-oriented sample			Full sample		
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8	Model 9
Firm age	.032 (.092)	.023 (.089)	.024 (.090)	-.146* (.070)	-.143* (.069)	-.131* (.063)	-.078 (.055)	-.077 (.055)	-.076 (.054)
Firm size	-.006 (.007)	-.005 (.006)	-.005 (.007)	.002 (.002)	.001 (.002)	.001 (.002)	.001 (.002)	.001 (.002)	.001 (.002)
Founding team size	.014 (.054)	.008 (.052)	.014 (.053)	.077 (.086)	.067 (.084)	.094 (.076)	.008 (.042)	.007 (.042)	.004 (.041)
R&D team size	-.003 (.015)	-.001 (.015)	-.001 (.015)	-.007 (.016)	.002 (.016)	.003 (.015)	-.006 (.010)	-.006 (.010)	-.003 (.010)
Marketing team size	-.026 (.038)	-.019 (.037)	-.024 (.038)	.005 (.015)	-.001 (.015)	-.017 (.014)	.004 (.014)	.004 (.014)	.001 (.014)
Investor involvement	.746 (.477)	.696 (.462)	.765 (.477)	-.300 (.333)	-.253 (.328)	-.359 (.295)	.204 (.259)	.211 (.260)	.213 (.253)
Founder age	.280 (.273)	.220 (.266)	.199 (.270)	.339 (.211)	.399† (.209)	.261 (.189)	.323† (.164)	.322† (.164)	.312† (.160)
Start-up experience	.254 (.417)	.395 (.408)	.350 (.421)	-.546† (.326)	-.766* (.334)	-.746* (.300)	-.314 (.246)	-.299 (.248)	-.348 (.242)
Learning experience	.275 (.357)	.232 (.347)	.211 (.355)	.408 (.275)	.364 (.271)	.350 (.243)	.389† (.214)	.388† (.215)	.327 (.210)
Institutional uncertainty	-.002 (.197)	.061 (.193)	.066 (.195)	.086 (.150)	.097 (.148)	.179 (.133)	.037 (.115)	.036 (.115)	.073 (.112)
Technological turbulence	.358† (.207)	.346† (.201)	.312 (.209)	-.075 (.129)	-.060 (.127)	-.030 (.116)	.069 (.109)	.067 (.109)	.078 (.106)
Technological openness	-.131 (.225)	-.188 (.219)	-.138 (.425)	.109 (.147)	.101 (.144)	-.230 (.204)	.016 (.119)	.015 (.119)	-.007 (.117)
User interactivity	.188 (.200)	.086 (.198)	.303 (.380)	-.439** (.134)	-.506*** (.135)	.289 (.197)	-.258* (.110)	-.253* (.111)	-.326** (.110)
Industry Control			Control	Control	Control	Control	Control	Control	Control
SOBMD		.745* (.286)	.742* (.293)		-.644* (.270)	-.949*** (.255)		.058 (.112)	-.038 (.112)
SOBMD ²									.333*** (.089)
SOBMD × Technological openness			-.013 (.364)			-.582* (.248)			
SOBMD × User interactivity			-.242 (.360)			1.155*** (.217)			
Constant	-1.022 (1.342)	-1.818 (1.337)	-1.617 (1.381)	-.207 (1.919)	-.433 (1.889)	.926 (1.715)	-.342 (1.865)	-.334 (1.868)	-.457 (1.821)
R ²	.178	.235	.239	.258	.288	.437	.111	.112	.160
Adjusted R ²	-.037	.025	.008	.104	.134	.305	.013	.010	.059
F value	.829	1.119	1.036	1.673*	1.867**	3.310***	1.128	1.095	1.586*

Note: SOBMD = strategic orientation in business model design; Standard errors are reported in the parentheses; †, *, **, and *** denote significance at the 10%, 5%, 1% and .1% levels.

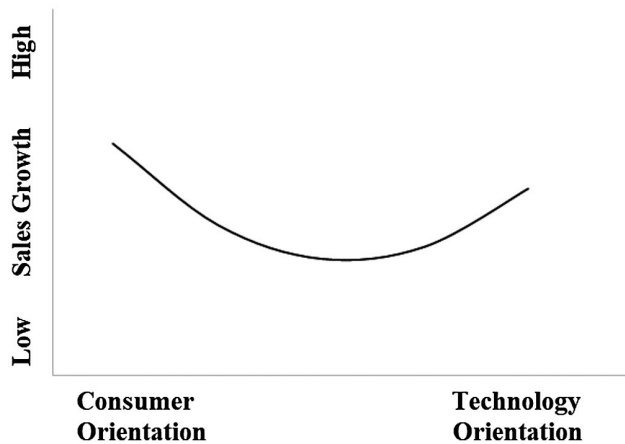


Figure 2. The Effect of Strategic Orientation in Business Model Design on Sales Growth

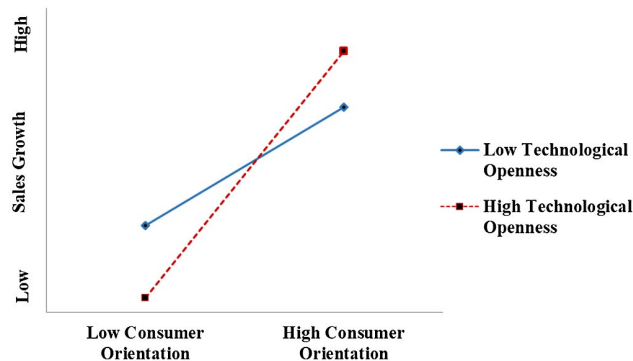


Figure 3. Consumer Orientation and Sales Growth: The Moderating Role of Technological Openness [Colour figure can be viewed at [wileyonlinelibrary.com](#)]

These results provide clear evidence for H1, H2, and H3. We then plotted the moderating effects of technological openness and user interactivity. For example, when technological openness is higher, the positive relationship between consumer-oriented business model design and sales growth becomes stronger (Figure 3), thus supporting H6. Taken together, these results provide additional support for our hypotheses (Figure 4).

Discussion

Employing a data set of 281 Chinese digital start-ups, this study finds that both technology and consumer orientations in business model design are beneficial to the performance of start-ups. However, a digital start-up's attempt to incorporate both strategic orientations

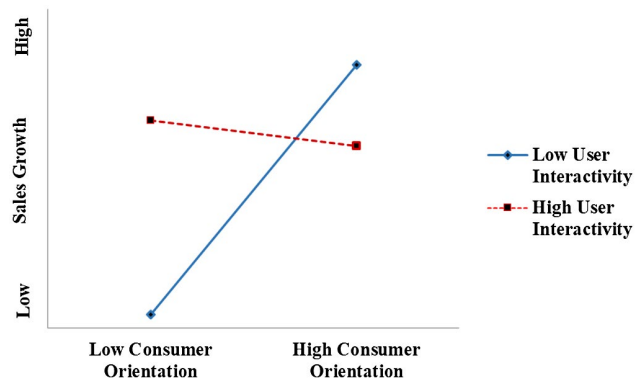


Figure 4. Consumer Orientation and Sales Growth: The Moderating Role of User Interactivity [Colour figure can be viewed at [wileyonlinelibrary.com](#)]

is counterproductive. That is, instead of seeking a balance between technology orientation and consumer orientation, the digital start-up should choose one when designing its business model. Furthermore, we find that technological openness strengthens the positive relationship between consumer-oriented business model design and the performance of start-ups, but user interactivity weakens this relationship. However, our hypotheses regarding the moderating effects of a digital environment on the relationship between technology-oriented business model design and the performance of start-ups are not supported by the data, suggesting that the performance implications of technology-oriented business model design are less sensitive to the borderless digital environment.

Theoretical Implications

This study makes two major contributions. First, it proposes and tests a strategic orientation view of business model design in the context of the digital economy. Regarding business model design, there exist at least two streams of research. The evolutionary view focuses on the process of business model design, by addressing the emergence (McDonald and Eisenhardt, 2020), innovation (Snihur and Zott, 2020), transformation (Aspara et al., 2013), or evolution (Demil and Lecocq, 2010) of the business model. Meanwhile, the thematic view focuses on the central idea of business model design, by emphasizing design themes or typologies of the business model (Amit and Zott, 2001; Kulins et al., 2016; Pati et al., 2018; Teece, 2010; Zott and Amit, 2007, 2008, 2010). Despite that, prior literature on business model design has largely

been limited to the context of the traditional economy or e-business (e.g., Amit and Zott, 2001).

This study diverges from existing schools of thought by advancing understanding of the role of strategic orientation for successful business model design in the context of the digital economy. Specifically, drawing on insights from strategic orientation literature (Gatignon and Xuereb, 1997; Hakala, 2011; Zhou et al., 2005), we propose two strategic orientations in the business model design of digital start-ups, technology orientation and consumer orientation. By finding that both orientations are beneficial for performance, we show that digital start-ups could adopt either technology or consumer orientation in business model design to boost their growth. However, our findings also suggest that, given the limited resources available to start-ups (De Massis et al., 2017; Zott and Amit, 2007) and conflicting business logics (Ross et al., 2017), digital start-ups should avoid seeking to strike a balance between technology orientation and consumer orientation. Further, our findings provide clear evidence for the alternative approach to understand the interaction of multiple strategic orientations (Hakala, 2011; Hortinha, Lages, and Lages, 2011). Overall, this study adds knowledge to business model design literature by proposing a strategic orientation view in the field of entrepreneurship and the context of the digital economy.

Second, this study contributes to business model theory building by introducing an integrative view that embraces both the RBV and the DSP. By drawing insights from both the RBV and the DSP, we examine the impact of both technology- and consumer-oriented business model design on start-up performance. Our results clearly show that both theories work well, in that either technology orientation or consumer orientation can significantly improve the start-up performance.

In particular, we contribute to an integrative view of the RBV and the DSP by disclosing the complexity of their integration. While recent scholars have called for equally addressing the resource and demand side (e.g., Amit and Han, 2017; Priem et al., 2013), our findings show that it is not that simple. Unlike Priem et al. (2013), our findings regarding the joint effect of technology orientation and consumer orientation shows the substitutability nature of the two theories. Specifically, our results indicate that attempts by start-ups to embrace both technology and consumer orientation in business model design concurrently could lead to conflicts of business philosophies and

suboptimal resource allocation (Ross et al., 2017). Conversely, consistent with Priem et al. (2013), our findings about the contingent value of digital business environment provide some evidence for the complementarity of the two theories. Our findings suggest that a consumer-oriented business model works better in highly open technological environments, by making up for the start-up's deficiency of technological resources. However, a consumer-oriented business model is less effective in highly interactive demand environments, largely due to the existence of the substitution effect. In short, our findings not only help reconcile different viewpoints from the RBV and the DSP, but also provide new insights into an integrative view of the RBV and the DSP.

Managerial Implications

This study further provides important implications for practitioners. First, our findings guide entrepreneurs to effectively design their business models in the digital economy. While both technology and consumer orientations work well for digital start-ups, balancing the two in business model design is detrimental to firm performance. Thus, we suggest that entrepreneurs subscribe to either a “technology push” or a “market pull” philosophy in business model design, but not both. Choosing the former entails a commitment to R&D, the acquisition of new technologies, the application of the latest technologies, and the introduction of new products or innovations. The latter, meanwhile, involves putting the customer's interest first and emphasizing the creation of superior customer value by identifying, analyzing, understanding, and answering customer needs. In addition, for those start-ups whose business models are market- or consumer-oriented, they should take full advantage of technological resources available in the open technological environment.

Second, our findings have implications for business incubators. By offering office space, providing services, and building connections, incubators can increase the chance of survival of start-up companies. How should business incubators strategically position themselves? Our findings suggest that, playing as an entrepreneurial ecosystem linking start-ups with the external environment, business incubators should specialize in either technology or the consumer market. Thus, we offer two suggestions for business incubators. First, they should choose one of the two alternative

development paths, in order to become either a technology-centered or a consumer-centered incubator. Second, we encourage business incubators to focus on one or a few areas or industries. If an incubator positions itself as technology-oriented, it should try to build a technology-rooted incubation ecosystem by entering into an area with great technology-originated entrepreneurial opportunities, attracting technology entrepreneurs in this area, and connecting technology ventures with each other as well as with various technical service providers. However, if an incubator positions itself as market-oriented, it should try to build a market-rooted incubation ecosystem by entering into an area with huge market-originated entrepreneurial opportunities, attracting consumer-facing entrepreneurs, and connecting them with various service providers that could help them better understand and exploit their markets.

Third, our findings offer suggestions for investors as well. The business model is a key factor for investment screening and evaluation. Similar to business incubators, we have two suggestions for venture capitalists and angel investors. First, we suggest that investors focus either on technology-oriented or consumer-oriented projects. Compared to consumer orientation, technology orientation carries greater risks and attaches greater importance to the firm's long-term success. For those investors who hold a "high risk and high return" philosophy, they can specialize in investing in technology start-ups in certain fields of technology or related industries. If they prefer investments that provide immediate returns, consumer-oriented projects might be a better option. Second, once an investor finds a clear investment positioning, they should back start-ups not only by funding them, but also by building networks among invested start-ups and external supporters, as well as offering professional suggestions for management decisions. Technology investors, for example, should back start-ups with their own technological knowledge or by enabling start-ups to gain access to external technological resources such as key technologies or technical talent.

Limitations and Future Research

We acknowledge several limitations of this study. First, our empirical results should be interpreted with caution due to the defect of measures. All scales for this study were developed based on existing literature and adapted to the context of the digital economy.

However, although our measures show acceptable reliability and validity, factors like strategic orientation in business model design are rather recent, in that there is not much experience to draw upon. Second, like many other studies, the cross-sectional data adopted in this study do not allow for causal analysis. Future studies should try to use panel data or longitudinal data to test the hypotheses. Finally, there is the problem of generalization. This study is posited in the Chinese context. Despite China's rapidly growing digital economy, economies from other areas such as the United States are also undergoing profound digital transformations. Our data set collected from one single economy prevents us from drawing universal conclusions for several reasons. For one, China's digital economy is much more market-driven as compared to that of the United States. That is why 59.6% of our sampling firms are categorized as consumer-oriented. A survey by Boston Consulting Group in 2017 revealed that only 10% of China's unicorn internet companies are completely technology-oriented. This sampling bias might be a threat to our findings. For another, unlike other countries, China's digital economy is strongly supported by the government. Following the regulation principle of "prudential tolerance," the Chinese central government encourages the rapid growth of indigenous technology-based digital ventures, leaving their sustainable growth an open issue. Thus, if we take the long-term growth of digital start-ups into account, our findings might become different.

Meanwhile, we see several avenues for future research. First, scholars should continue to explore the issue of how to achieve a balance between technology orientation and consumer orientation in business model design. Due to the conflicting logics and the competition for scarce resources, we argue that digital start-ups should adopt an alternative strategic orientation view. However, firms that have enough resources, such as large established firms, might be able to balance between the two strategic orientations (a balanced view) or even stand out in both strategic orientations (an ambidextrous view). Thus, further efforts should be made to figure out the applicable conditions for different views.

Second, comparative studies should be conducted to examine whether our theoretical model operates similarly in different contexts. For example, consumer sovereignty is given unprecedented attention in the digital economy, in that digital technologies shorten the distance between firms and their

consumers and make the power sources shift significantly to consumers (Labrecque, vor dem Esche, Mathwick, Novak, and Hofacker, 2013). Therefore, it is worth speculating whether the consumer-oriented business model plays a more important role in the digital economy as compared to the traditional economy. Furthermore, the digital economy reveals different features in different countries. As previously mentioned, the digital economy of China is primarily market-driven, but that of the United States is much more technology-driven. Thus, it becomes necessary to examine how technology- and demand-driven business model designs impact firm performance in different countries.

Finally, the design of some particular type of business models deserves attention in the digital economy. For example, in recent decades, we have witnessed a proliferation of digital platforms such as Facebook, YouTube, Twitter, WeChat, among others. On the one hand, representing an open business model, the success of a digital platform requires support from complementary innovators (Srinivasan and Venkatraman, 2018). On the other hand, the digital platform enables start-ups to pursue entrepreneurial opportunities by creating complementary products or services (Eckhardt, Ciuchta, and Carpenter, 2018). Therefore, how to design a platform business model becomes an interesting proposition to explore.

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Appendix A. Scales and Items for Strategic Orientation in Business Model Design

Consumer orientation	Loading	VIF	Technology orientation
<i>Value creation</i>			
The greatest advantage of my firm is to identify and exploit business opportunities	.774***	1.271	The greatest advantage of my firm is in technological innovation
We provide industry-leading insights about consumer needs	.681***	1.427	We are in an industry-leading position in mastering advanced knowledge and technical expertise
We often employ mature and stable technologies, which is a direct application of existing technologies	.840***	1.317	We often employ advanced and innovative technologies, which makes a disruption or a big challenge to existing technological system
<i>Value proposition</i>			
The idea of product or service comes from consumer needs	.918***	1.840	The idea of product or service comes from the support of some unique technology
We are always consumer- or user-oriented	.835***	1.759	We are always technology- or product-oriented
Users choose our product or service that can best meet their needs	.773***	1.710	Users choose our product or service because of the advantage and reliability of our technologies
<i>Value capture</i>			
The key to our profitability is strict cost control	.892***	1.153	The key to our profitability is heavy R&D investment
The product or service we offered often should be paid, because it is our key source of revenue	.746***	1.153	The product or service we offered can be free, and we often profit from secondary development of our primary business

Note: All variables were measured by a 5-point scale, 1 indicating strongly agree with demand-driven business model design, 2 agree with demand-driven, 3 neutral, 4 agree with technology-driven, and 5 strongly agree with technology-driven; *** denotes significance at the .1% level.

Appendix B. Scales and Items for Technological Openness and User Interactivity

Item	Loading
<i>Technological openness</i> ($\alpha = .845$; $AVE = .613$; $C.R. = .888$)	
Many technologies we rely on are free of charge	.725
We can easily access technical information in our field	.791
There are a lot of technical talents in our field	.775
It is easy to find technical service providers in our field	.795
We can get technical services at a low cost	.824
<i>User interactivity</i> ($\alpha = .798$; $AVE = .713$; $C.R. = .882$)	
Consumers can interact with us frequently	.849
Consumers can interact with us instantly	.873
Consumers can communicate with us through various channels, including online communities, social networking services, and apps	.810

Note: All variables were measured by a 5-point scale ranging from “strongly disagree” to “strongly agree.”